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STATE-OF-THE-ART

Why do people participate in Web surveys? Applying survey participation theory to Internet survey data collection

Florian Keusch

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Abstract In recent years Web surveys have emerged as the most popular mode of primary data collection in market and social research. To improve our understanding about the influence of different societal-level factors, characteristics of the sample person, and attributes of the survey design on participation in Web surveys, this paper establishes a systematic link between theoretical frameworks used to explain survey participation behavior and state-of-the-art empirical research on online data collection methods. The concepts of self-perception, cognitive dissonance, commitment and involvement, social exchange, compliance, leverage-salience, and planned behavior are discussed and their relationship with factors that have empirically proven to influence Web survey participation are analyzed using data from an expert survey. This paper will help researchers and practitioners to make informed decisions about the use of techniques increasing participation in Web surveys.

Keywords Empirical market and social research · Survey methodology · Web survey participation behavior · Self-perception theory · Cognitive dissonance theory · Commitment · Involvement · Social exchange theory · Compliance heuristics · Leverage-salience theory · Theory of planned behavior · Expert survey · Principal component analysis

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F. Keusch (✉)
Lehrstuhl für Statistik und sozialwissenschaftliche Methodenlehre, Universität Mannheim, 68131 Mannheim, Germany
e-mail: f.keusch@uni-mannheim.de

1 Introduction

Gaining cooperation to participate in surveys is essential to the success of all primary data collection in market and social research. Low response rates potentially introduce nonresponse bias, in particular if respondents and nonrespondents differ in the variable of interest (Groves 2006), jeopardizing the validity of survey results and their managerial implications. Declining response rates and the challenge of nonresponse have been unresolved issues in survey research for decades now (Curtin et al. 2005; de Leeuw and de Heer 2002; Steeth 1981). In the meantime, the rise of the Internet has dramatically influenced both people's daily lives and how scientists conduct research (Denissen et al. 2010). More than ever before, managerial decisions rely on data collected through self-administered Web surveys where consumers fill out online questionnaires on their computers, laptops, or mobile devices through an Internet connection. ESOMAR (2013) estimates that Web surveys account for 27 % of the total market research investment worldwide (including both qualitative and quantitative methods), well ahead of telephone surveys (13 %) and face-to-face surveys (12 %).

Although Web surveys are a cost- and time-efficient alternative compared with the traditional modes of data collection (Evans and Mathur 2005), the introduction of Web surveys has not solved the problem of declining response rates. On the contrary, several meta-analyses show that traditional survey data collection modes outperform Web surveys in terms of their response rate (Cho et al. 2013; Lozar Manfreda et al. 2008; Shih and Fan 2008; Yarger et al. 2013). Thus, "nonresponse bias represents an even more threatening factor for the validity of results obtained via Web surveys" (Bosnjak et al. 2005, p. 490). Survey researchers must understand the psychological mechanisms underlying the survey participation process to make informed decisions about the use of techniques thought to influence participation behavior in Web surveys.

Research on participation in surveys has mainly focused on examining design factors of surveys, such as characteristics of the participation request, questionnaire design, survey sponsor, and survey topic. A number of quantitative reviews analyze the influence of various factors on participation behavior in different data collection modes (see meta-analyses by Bruvold et al. 1990; Cho et al. 2013; Church 1993; Cook et al. 2000; Edwards et al. 2002, 2009; Fox et al. 1988; Göritz 2006a; Kanuk and Berenson 1975; Lozar Manfreda and Vehovar 2002; Lozar Manfreda et al. 2008; Shih and Fan 2007, 2008; Yarger et al. 2013; Yu and Cooper 1983). However, "any reading of this large body of works leads to the conclusion that few survey design factors have a consistent and significant effect on observed response rates" (Helgeson et al. 2002, p. 305) and "the majority of these studies have made no attempt at building theory or explaining response behavior using a theoretical model" (Cavusgil and Elvey-Kirk 1998, p. 1165). Albaum et al. (1998) suggest drawing a clear line between theoretical explanations for survey participation behavior and techniques used to increase survey response. Although the two concepts are inextricably linked to each other, a clear demarcation is of paramount importance because theory helps to understand how these factors influence the survey participation decision and provides guidance to researchers for survey design. Only during the last two decades, theoretical models emerged that explain participation behavior in surveys. The fact that most of these theories are based in traditional data collection modes, i.e., personal, telephone, and mail surveys, calls for their reevaluation under Web survey conditions.

The main purpose of this paper is to improve the understanding about the influence of different factors when inviting people to participate in Web surveys by linking existing theoretical frameworks on participation behavior to state-of-the-art empirical results from methodological research on Web surveys. I will start by giving a systematic review of the scientific literature on the influence of different factors on participation behavior in Web surveys. As the number of empirical studies in this area has increased dramatically over the last years, this section of the paper provides an update of an earlier review by Fan and Yan (2010). Next, this paper discusses theoretical frameworks used in the literature to explain participation behavior in surveys in general and how these models apply to the idiosyncrasies of Web surveys. Third, I present the results from an expert survey that links the theoretical survey participation frameworks to factors influencing participation behavior in Web surveys. I conclude with a discussion of the findings and suggestions for future research to cover unchartered territories.

2 Factors affecting response behavior in Web surveys

Groves et al. (1992) identify three categories of factors that influence participation in self-administered surveys: (1) societal-level factors, (2) characteristics of the sample person, and (3) attributes of the survey design. Based on published studies in peer-reviewed journals found through an extensive literature search, I will give a systematic review of different factors affecting response behavior in Web surveys in these three categories. To locate scientific literature, I searched major electronic reference databases (Cooper 2010; Reed and Baxter 1994), including ERIC, JSTOR, MEDLINE, PsycARTICLES, PsycINFO, Pubmed, Scopus, Science Direct, Social Science Abstracts, Sociological Abstracts, Web of Science: Social Science Citation Index, WebSM, and Wilson OmniFile Full Text Select. Following the procedure suggested in an earlier review on factors influencing participation in Web surveys (Fan and Yan 2010), I used three groups of key words for the search: (a) *Web*, *www*, *Internet*, *Web-based*, *online*, *electronic*; (b) *questionnaire*, *survey*, *data collection*; (c) *response rate*, *return rate*, *participation rate*. Additionally, I manually searched for references included in the literature found through the database search. I stopped the literature search on April 3rd, 2014, and any study published after this date is not included in this review. Table 1 summarizes the empirical findings on factors influencing participation behavior in Web surveys.

2.1 Societal-level factors

Factors that influence survey participation on a society level include the felt legitimacy of organizations collecting survey data, the degree of social cohesion, and survey fatigue caused by over-surveying people (Groves et al. 1992). Only a few studies have empirically analyzed the influence of societal-level factors on survey participation in general and in Web surveys in particular. Porter et al. (2004) demonstrate that survey fatigue exists among university students invited to multiple Web surveys.

Table 1 Summary of empirical findings on factors influencing participation behavior in Web surveys

	Studies	Findings
Societal-level factors		
Survey fatigue	Porter et al. (2004)	More Web survey invitations lead to lower participation
Culture	Petrova et al. (2007)	Stronger orientation toward collectivism increases participation
Sample person characteristics		
Gender	Busby and Yoshida (2013), Dykema et al. (2012), Laguilles et al. (2011), Marcus and Schütz (2005), Patrick et al. (2013), Porter and Whitcomb (2005b)	Females are more likely to participate than males
Race/ethnicity	Patrick et al. (2013)	Blacks are less likely to participate than non-blacks
Population	Boulianne et al. (2011), Kaplowitz et al. (2012), Mavletova et al. (2014), Shih and Fan (2008)	Mixed findings for student vs. non-student populations
Personality	Brüggen and Dholakia (2010), Fang et al. (2009), Marcus and Schütz (2005), Petrova et al. (2007), Porter and Whitcomb (2005b), Stingelbauer et al. (2011)	Web survey respondents show more social engagement, investigative personality, openness to experience, need for cognition, curiosity, agreeableness, and Web innovativeness and less artistic personality, enterprise personality, and conscientiousness than nonrespondents
Personal topic interest	Edwards et al. (2009), Galesic (2006), Haunberger (2011), Keusch (2013), Marcus et al. (2007), McCambridge et al. (2011), Porter and Whitcomb (2005a), Postoaca (2006), Tourangeau et al. (2009), Zillman et al. (2014)	Personal interest in survey topic influences participation decision in cross sectional Web surveys
Attitudes towards survey research	Bosnjak et al. (2005), Brüggen et al. (2011), Haunberger (2011), Keusch et al. (2014), Peytchev (2011)	Positive attitudes toward surveys in general lead to higher Web survey participation
Previous participation behavior	Göritz (2008, 2014), Haunberger (2011), Keusch (2013), Petrova et al. (2007), Peytchev (2011), Svensson et al. (2012)	Individuals show consistency in (non)participation behavior across several waves of Web surveys and in online panels
Survey design attributes		
Contact		
Contact mode	Birnholtz et al. (2004), Dykema et al. (2012), Kaplowitz et al. (2012), Millar and Dillman (2011), O'Neil et al. (2003), Schillewaert and Meulemeester (2005)	Mixed results for invitations sent via mail vs. e-mail; samples drawn from online panels more likely to participate than samples drawn from other online sources

Table 1 continued

	Studies	Findings
Prenotification	Bandilla et al. (2012), Bosnjak et al. (2008), Dykema et al. (2011), Felix et al. (2011), Hart et al. (2009), Kaplowitz et al. (2004), Kent and Brandal (2003), Keusch (2012), Lozar Manfreda and Vehovar (2002), Porter and Whitcomb (2007), Wiley et al. (2009)	Prenotification via offline media enhances response rates in Web surveys; mixed findings for effect of e-mail prenotifications
Timing of invitation	Faught et al. (2004), Göritz (2014), Sauermann and Roach (2013)	Invitations sent on Wednesday morning yield highest response rates; invitations sent on weekend are not immediately returned; higher Web survey participation in winter
Sender/sponsor	Bosnjak and Batinic (1999), Boulianne et al. (2011), Fang et al. (2009, 2012), Guéguen (2003), Guéguen et al. (2005), Guéguen and Jacob (2002a, b), Joinson and Reips (2007), Joinson et al. (2007), Pan et al. (2013), Keusch (2012), Porter and Whitcomb (2003b, 2007), Sutherland et al. (2013), Tuten (1997), Tourangeau et al. (2009), Wiley et al. (2009)	Relationship between invitation sender and recipient, familiarity of sponsor, trust in sponsor, reputation of sponsor and survey provider, and authority of sender influence participation decision; mixed findings for academic vs. non-academic sender/sponsor; female senders increase participation in male populations
Subject line	Edwards et al. (2009), Kent and Brandal (2003), Kaplowitz et al. (2012), Keusch (2013), Mavletova et al. (2014), Porter and Whitcomb (2005a), Trouteaud (2004), Tuten (1997)	Promoting incentives and references to survey have negative effect while appeals for help and referencing an authority figure have positive effect on participation
Invitation message	Cook et al. (2000), Crawford et al. (2001), Edwards et al. (2009), Greif and Batinic (2007), Göritz and Stieger (2009), Heerwegh (2005), Heerwegh and Loosveldt (2002, 2003, 2006, 2007), Heerwegh et al. (2005), Joinson and Reips (2007), Joinson et al. (2007), Kaplowitz et al. (2012), Kent and Brandal (2003), Klofstad et al. (2008), Mavletova et al. (2014), Messer and Dillman (2011), Porter and Whitcomb (2003b), Sánchez-Fernández et al. (2012), Sauermann and Roach (2013), Whitcomb and Porter (2004), Wiley et al. (2009)	Mixed findings for personalization of invitation, length of message, and whether or not deadline for participation is stated; having to type in password upon questionnaire access reduces willingness to participate

Table 1 continued

	Studies	Findings
Reminder	Batinic and Moser (2005), Burgesse et al. (2012), Göritz (2014), Göritz and Crutzen (2012), Keusch (2012), Klofstad et al. (2008), Lozar Manfreda et al. (2008), Miksza et al. (2010), Porter and Whitcomb (2007), Sánchez-Fernández et al. (2012), Sauermann and Roach (2013), Shih and Fan (2008), Wiley et al. (2009)	Sending e-mail reminders increases participation but reminders reach an early saturation point; postal reminders do not increase participation in Web surveys
Incentives		
Unconditional incentives	Birnholtz et al. (2004), Bosnjak and Tuten (2003), Dykema et al. (2011, 2012), Gajic et al. (2012), Göritz (2005, 2006a, 2008), Göritz et al. (2008), LaRose and Tsai (2014), Messer and Dillman (2011), Millar and Dillman (2011), Parsons and Manierre (2014), Patrick et al. (2013), Sánchez-Fernández et al. (2010)	Unconditional incentives increases participation; monetary incentives sent via PayPal do not affect participation
Conditional incentives	Busby and Yoshida (2013), Cobanoglu and Cobanoglu (2003), Doerflinger et al. (2010), Dykema et al. (2011), Gajic et al. (2012), Galesic (2006), Göritz (2004b, 2006a, b, 2008, 2014), Göritz and Luthe (2013a, b, c), Heerwegh (2006), Laguilles et al. (2011), Marcus et al. (2007), McCree-Hale et al. (2010), O'Neil and Penrod (2001), Göritz and Wolff (2007), O'Neil et al. (2003), Preece et al. (2010), Porter and Whitcomb (2003a, 2004), Sánchez-Fernández et al. (2012), Sauermann and Roach (2013), Wiley et al. (2009), Wilson et al. (2010), Ziegenfuss et al. (2013)	Mixed findings for incentives conditional upon finishing Web surveys
Survey results	Batinic and Moser (2005), Edwards et al. (2009), Galesic (2006), Göritz (2014), Göritz and Luthe (2013a, c), Marcus et al. (2007), Sax et al. (2003), Tuten et al. (2004)	Offering survey results as incentives does not increase participation
Questionnaire design		
Paging vs. scrolling	Lozar Manfreda et al. (2002), Peytchev et al. (2006)	No difference in participation between paging and scrolling format

Table 1 continued

	Studies	Findings
Progress indicator	Crawford et al. (2001), Conrad et al. (2010), Healy et al. (2005), Heerwegh and Loosveldt (2006), Matzat et al. (2009), Villar et al. (2013), Yan et al. (2011)	Static progress indicators have no or negative effect on survey completion; fast-to-slow progress indicators decrease and slow-to-fast progress indicators increase break-offs
Questionnaire layout	Göritz and Stieger (2008), Keusch (2012), Mahon-Haft and Dillman (2010), Stieger et al. (2011), Wiley et al. (2009)	Layout features of the Web questionnaire have no effect on survey completion; java applets and higher loading times lower likelihood of participation
Questionnaire length	Bosnjak and Batinic (1999), Crawford et al. (2001), Deutskens et al. (2004), Edwards et al. (2009), Galesic (2006), Galesic and Bosnjak (2009), Ganassali (2008), Göritz (2014), Kaplowitz et al. (2012), Marcus et al. (2007), McCambridge et al. (2011), Trouteaud (2004), Waltson et al. (2006), Wiley et al. (2009), Yan et al. (2011)	Actual length and announced length of Web questionnaire negatively correlated with participation behavior

Petrova et al. (2007) report a higher response rate to a Web survey in cultures with a stronger orientation toward collectivism, such as in Asia, compared to cultures with a stronger orientation toward individualism, such as in the US.

2.2 Characteristics of the sample person

According to Groves et al. (1992), sociodemographic factors are the most widely discussed sample person characteristics influencing survey participation. In addition, Fan and Yan (2010) suggest measuring when and how personality characteristics influence Web survey participation. Although Internet penetration is on the rise worldwide, gaps between different sociodemographic groups in terms of their access to the Internet still exist (see European Commission 2012 and Pew Research Center 2014 for European and US data, respectively), potentially causing coverage error when collecting survey data over the Internet. Hence, when analyzing the influence of sample member characteristics on Web survey participation, researchers must consider that their analysis is limited to people with access to the Internet.

Gender seems to be one of the strongest predictors for Web survey participation with several studies showing that males are less likely to participate than females (Busby and Yoshida 2013; Dykema et al. 2012; Laguilles et al. 2011; Marcus and Schütz 2005; Patrick et al. 2013; Porter and Whitcomb 2005b). Patrick et al. (2013) find that Blacks are less likely to participate in a Web survey compared with Non-blacks. Shih and Fan (2008) show in their meta-analysis that Web surveys among college students lead to higher response rates than among other populations, such as professionals, employees, and the general public. On the other hand, more recent studies by Boulianne et al. (2011) and Kaplowitz et al. (2012) find that the response rate to campus wide Web surveys is significantly higher among faculty and staff compared with students. Mavletova et al. (2014) find that faculty and staff were significantly more likely to start but also significantly more likely to break off from a Web survey than were students, resulting in no difference in overall response rates.

Looking at differences in personality traits, studies find that respondents to Web surveys show more social engagement, investigative personality (Porter and Whitcomb 2005b), openness to experience (Brüggen and Dholakia 2010; Marcus and Schütz 2005), need for cognition, curiosity, and agreeableness (Brüggen and Dholakia 2010), and less artistic personality, enterprise personality (Porter and Whitcomb 2005b), and conscientiousness (Marcus and Schütz 2005) than nonrespondents. Petrova et al. (2007) report that a strong personal orientation toward individualism increases the likelihood of participating in a subsequent Web survey compared with a strong orientation toward collectivism. Fang et al. (2009) find that personal Web innovativeness, i.e., a person's willingness to try out new Web services, correlates with Web survey participation intention. Stingelbauer et al. (2011) show that online panel members who have low trust in the anonymity of a Web survey are more likely to show enduring survey participation if they receive extrinsic or intrinsic motivation.

A number of studies find that sample members' personal interest in the survey topic influences their participation decision in cross-sectional Web surveys (Edwards et al. 2009; Galesic 2006; Haunberger 2011; Lozar Manfreda et al. 2002; Marcus et al. 2007; McCambridge et al. 2011; Porter and Whitcomb 2005a; Zillman et al. 2014). Membership surveys in online panels confirm that topic is one of the most crucial factors when deciding on survey participation (Postoaca 2006). However, experimental studies find inconclusive results about the influence of topic interest on participation in online panel Web surveys (Keusch 2013; Tourangeau et al. 2009).

Recent research on participation motives and behavior in online panels shows that positive attitudes toward surveys in general lead to higher Web survey participation (Bosnjak et al. 2005; Brüggen et al. 2011; Haunberger 2011; Keusch et al. 2014). Additionally, several studies find that individuals show consistency in their (non)participation behavior across Web surveys over several waves (Göritz 2008; Haunberger 2011; Petrova et al. 2007; Peytchev 2011; Svensson et al. 2012) and in multiple surveys in an online panel (Göritz 2014; Keusch 2013).

2.3 Attributes of the survey design

Design choices of surveys can have an impact on a sample member's participation decision in a variety of ways (Groves et al. 1992). Before considering individual attributes of survey design that effect participation, it is important to note how the design of Web surveys is similar to other data collection methods and how it differs. For example, like mail surveys, Web surveys are self-administered, meaning that there is no live interviewer present during the survey participation request and the completion of the questionnaire (Evans and Mathur 2005). Therefore, Web surveys are less limited by time restrictions of interviewers, and respondents can fill out the questionnaires at their own convenience. However, ignoring an e-mail invitation to a Web survey is easier than ignoring a personal invitation over the phone or face-to-face. On the other hand, Web questionnaires are similar to other data collection methods because they are written in programs similar to those used for computer assisted telephone and personal interviewing (Batinic 2002). This facilitates the use of multiple communication channels (i.e., aural, visual, oral) and interactive questionnaire features, such as automated skips, adaptive questioning, branching, and item randomization, and allows the researcher to track respondents' behavior during the entire survey participation process (Lozar Manfreda et al. 2002). The main differences between Web surveys and offline data collection methods stem from the virtual character of the Internet. This influences (1) how researchers can contact sample members to participate in Web surveys, (2) how incentives are used as inducement strategies, and (3) how respondents perceive the questionnaire. Each of these design aspects and their effect on response behavior in Web surveys will be considered in turn.

2.3.1 Contact

Contact mode Web questionnaires are stored on a Web server and individuals need to get access to the questionnaire online to participate. Thus, sample members must know the virtual address (i.e., the uniform resource locator, URL) of the server. Researchers have several options of informing sample members about the location of the Web questionnaire (Couper 2000). Web surveys where no sampling frame is available, such as surveys advertised through online banners (Schillewaert et al. 1998; Tuten et al. 2000) or offline media (Schillewaert et al. 1998), and pop-up intercept surveys (Comley 2000), usually yield extremely low response rates and suffer from a vague definition of the population they represent (Couper 2000). In well-defined populations where an exhaustive list of all population elements exists (e.g., students of a university, employees of a company, members of a professional organization), invitations to Web surveys are sent via e-mail including a clickable URL, or, if street addresses are available, via postal mail including a printed URL that the respondent has to type into a Web browser (Messer and Dillman 2011). Birnholtz et al. (2004) and Millar and Dillman (2011) find no significant difference in the response rate between Web surveys that use mail and e-mail invitations. Dykema et al. (2012) report that a mailed invitation works better in recruiting university faculty to a Web survey compared to an e-mail invitation. On the contrary, in the study by Kaplowitz et al. (2012) a postcard invitation is less successful than an e-mail invitation in recruiting faculty and staff to a campus wide university Web survey while there is no significant difference between e-mail and postcard invitation for students.

Using e-mail addresses of individuals with whom the survey sponsor has no prior established relationship (i.e., cold addresses) is seen as a violation of netiquette and even illegal in most countries, thus considered an unacceptable survey practice (Dillman et al. 2009). Therefore, commercial and non-profit research organizations have turned to pre-recruiting consumers into online panels. By opting in for an online panel, willing persons give their consent to regularly receive invitations for Web surveys from the online panel provider on a variety of topics (Evans and Mathur 2005; Göritz 2004a). Samples drawn from online panels have proven to produce fewer break-offs (O'Neil et al. 2003) and higher participation rates (Schillewaert and Meulemeester 2005) than have other online recruiting techniques.

Prenotification In contrast to follow-up reminders, which are very popular for mail and Web surveys alike, many Web surveys lack prenotification messages and the empirical results on their response enhancing effect are rather inconclusive. Three studies demonstrate a positive effect of e-mail prenotification on participation behavior (Keusch 2012; Lozar Manfreda and Vehovar 2002; Wiley et al. 2009) while three do not (Felix et al. 2011; Hart et al. 2009; Kent and Brandal 2003). Prenotification by means of other media, such as mailed letters or postcards (Bandilla et al. 2012; Dykema et al. 2011; Kaplowitz et al. 2004; exception: Porter and Whitcomb 2007) and mobile text messages (Bosnjak et al. 2008) consistently prove to be effective techniques to enhance response rates in Web surveys.

Timing of the invitation Faught et al. (2004) find in an experiment sending e-mail invitations over 14 time points in one week that the invitation sent on Wednesday morning yields the highest response rate. While Sauermann and Roach (2013) do not find a significant difference in response rates to a Web survey across the days of the week that the e-mail invitation is sent, they do report that invitees are significantly less likely to respond on the same day, if the invitation comes on the weekend. Göritz (2014) reports for an online panel that panel members are more likely to start and finish studies in winter than during any other season.

Sender and survey sponsor Compared with mailed pieces that have the advantage of numerous visual cues (e.g., size and color of envelope, special issue stamp) that could attract invitees to open and assess a survey request, electronic mail offers only limited features to induce the recipient to open the e-mail (Tuten 1997). The name and the e-mail address of the sender, together with the subject line, are the only pieces of information the recipient of an e-mail can use to evaluate the message without actually opening it. Tuten (1997) shows that e-mail recipients are in general more likely to open messages that come from individuals whose names are recognized. In the context of Web surveys, several studies report that the relationship between the sender and the invitation recipient (Porter and Whitcomb 2007), the familiarity of the sponsor (Pan et al. 2013), the trust in the sponsor (Fang et al. 2009), and the reputation of the sponsor and the survey provider (Fang et al. 2012) influence the participation decision. Only Tourangeau et al. (2009) cannot confirm that the affinity with the sponsor influences the Web survey participation decision of online panel members.

Several Web surveys find that the authority of the sender expressed through the signature in the e-mail invitation has a positive influence on the response rate (Guéguen and Jacob 2002b; Joinson and Reips 2007; Joinson et al. 2007; Pan et al. 2013; Sutherland et al. 2013; exception: Porter and Whitcomb 2003b). Boulianne et al. (2011) show no significant difference in the response rate between a campus-wide transportation survey that comes from the university's survey research center and the university's transportation department.

Bosnjak and Batinic (1999) find that academic e-mail surveys are perceived by respondents as being better and more positive, interesting, and likeable than surveys from commercial organizations. However, a more recent experimental study shows no influence of the sender's and sponsor's affiliation with a university on Web survey participation (Wiley et al. 2009).

Guéguen (2003) and Guéguen et al. (2005) report that if the invitation to a student Web survey comes from peers who have the same first or last name as the invitee, the likelihood of filling out the survey is higher than if the survey participation request comes from someone with a different name. Guéguen and Jacob (2002a) also find that the inclusion of a photo of the requester in the invitation e-mail to a student Web survey increases the response rate over no photo. Both Guéguen and Jacob (2002a) and Keusch (2012) show that the response probability to a Web survey among male respondents is significantly higher, if the invitation comes from a female sender.

Subject line Porter and Whitcomb (2005a) see e-mail subject lines as the equivalent of postage stamps or envelope appearance in mail surveys. Tuten (1997) shows that e-mail recipients are more likely to open messages with topics of personal interest in the subject line. Empirical research on subject line text in e-mail invitations for Web surveys finds that promoting incentives for participation (Kent and Brandal 2003) and references to the survey (Edwards et al. 2009; Porter and Whitcomb 2005a) have a negative effect on participation propensity while appeals for help (Trouteaud 2004; Porter and Whitcomb 2005a; exception: Mavletova et al. 2014) and referencing the request to participate from an authority figure (Kaplowitz et al. 2012) increase participation. Naming the survey topic in the e-mail invitation to a Web survey in an online panel has no effect on response rate (Keusch 2013).

Invitation message In contrast to mailed invitation packages that include both a cover letter and the questionnaire, survey requests via e-mail solely rely on the text that is sent with the electronic message. Fan and Yan (2010) identify three issues of the e-mail invitation message that researchers have extensively studied: (1) personalization of the e-mail invitation, (2) content of the e-mail invitation, and (3) access to the Web questionnaire. In terms of the personalization of the e-mail invitation to Web surveys, i.e., addressing sample members by name, the literature is rather inconclusive. While one meta-analysis (Cook et al. 2000) and seven individual studies find a positive effect of e-mail personalization (Heerwegh 2005; Heerwegh and Loosveldt 2006, 2007; Heerwegh et al. 2005; Joinson and Reips 2007; Sánchez-Fernández et al. 2012; Sauermann and Roach 2013) four others do not (Joinson et al. 2007; Kent and Brandal 2003; Porter and Whitcomb 2003b; Wiley et al. 2009).

Also the literature on the effect of different components of the content of the e-mail invitation to Web surveys, such as the length of the e-mail invitation message (Greif and Batinic 2007; Kaplowitz et al. 2012; Klofstad et al. 2008; Mavletova et al. 2014) and whether or not a deadline for the participation is stated (Edwards et al. 2009; Göritz and Stieger 2009; Porter and Whitcomb 2003b), on participation behavior in Web surveys is rather inconclusive. Felix et al. (2011) show that the use of a pleading appeal in the e-mail invitation does not significantly increase the response to a Web survey among academics authors. Keusch (2013) finds that fresh panel members have a higher participation rate when the topic is announced in the e-mail invitation whereas online panel members with longer membership tenure are not affected by revealing the topic in advance. Whitcomb and Porter (2004) report that while a simple HTML e-mail invitation outperforms a text e-mail, black background color and a complex header decrease response propensity.

Password protecting the access to the Web questionnaire allows survey designers to identify individual sample members based on their participation behavior, to provide a link between frame data and responses, and to safeguard the survey against unwanted access from people outside the sample or multiple participations from one same sample member. Providing a password that sample members have to type in upon questionnaire access might be perceived as too burdensome and therefore reduce the willingness to participate (Crawford et al. 2001; Heerwegh and Loosveldt 2002, 2003; Joinson et al. 2007). Hence, Web survey invitations via e-mail usually include an individualized link to the Web questionnaire that already contains the unique password (i.e., identifier key technique) so that no additional password entry is needed. Kaplowitz et al. (2012) show that placing the URL at the bottom of the e-mail invitation increases the response rate over placing the URL on the top. Messer and Dillman (2011) find that including a Web card to the postal mail invitation package that explains the respondents how to access the Web questionnaire does not improve the response rate over no Web card.

Reminder Due to minimal additional costs and low administrative effort, sending out e-mail reminders to individuals who have not reacted to the initial survey invitation is very tempting. Results from a meta-analysis (Shih and Fan 2008) and individual studies (Batinic and Moser 2005; Burgesse et al. 2012; Göritz 2014; Göritz and Crutzen 2012; Keusch 2012; Klofstad et al. 2008; Sauermann and Roach 2013; Wiley et al. 2009) demonstrate that follow-up reminder e-mails sent to nonrespondents have a positive effect on Web survey participation. However, Lozar Manfreda et al. (2008) find in their meta-analysis that reminders in Web surveys reach an earlier saturation point compared to other survey modes. A more recent experiment by Sánchez-Fernández et al. (2012) shows no increase in response rate of five over three reminders.

The postal follow-up letter mailed by Miksza et al. (2010) does not increase the response rate to a Web survey over no follow-up. Porter and Whitcomb (2007) find no significant difference in response rates between a postal and an e-mail reminder. Sauermann and Roach (2013) report no difference in response rates when sending reminders with different delays after the initial invitation. However, changing the wording of the subsequent reminder instead of using the same wording has a significant positive effect on response rates.

2.3.2 Incentives

The influence of incentives on response rates is among the most intensively studied subjects in survey methodology (see Singer and Ye 2013 for an extensive review), with research consistently finding stronger effects for unconditional incentives, i.e., pre-paid monetary token provided to all sample members in advance, over incentives conditional upon survey participation (Church 1993; Edwards et al. 2002, 2009). Göritz (2006a) finds in two meta-analyses incorporating a total of 32 studies, that incentives can have a positive influence on response and break-off rate in Web surveys. However, the virtual and intangible character of the Internet raises problems when trying to administer cash incentives with e-mail invitations (Deutskens et al. 2004).

Unconditional incentives In Web surveys where sample members are contacted via mail or in person, unconditional cash incentives can be administered together with the survey request. Several studies confirm for Web surveys that adding pre-paid incentives to the postal mail invitation increases the response rate over other strategies, such as sending just the survey invitation letter without an incentive (Messer and Dillman 2011; Millar and Dillman 2011; Parsons and Manierre 2014; exceptions: Dykema et al. 2011, 2012), providing an online gift card (Birnholtz et al. 2004), or offering a conditional lottery cash incentives (Gajic et al. 2012; LaRose and Tsai 2014). Patrick et al. (2013) find that a \$2 pre-paid incentive in combination with a \$10 post-paid incentive conditional on survey completion increases the response rate to a student Web survey over a \$10 pre-paid incentive. Boulianne (2012) finds that female students are more likely to respond to a mailed Web survey invitation that includes a \$5 pre-paid cash incentive compared to a \$10 pre-paid incentive. For male students the difference in response rates between the two incentives is not significant. Bosnjak and Tuten (2003) show that a monetary incentive sent via PayPal along with the e-mail invitation to a Web survey does not increase the response rate over no incentive and Göritz et al. (2008) even find a negative effect of a PayPal incentive. Also non-monetary incentives can increase the response rate, although to a lesser degree than unconditional cash incentives and only as long as they are provided unconditionally to all sample members (Göritz 2005, 2008; Sánchez-Fernández et al. 2010).

Conditional incentives Although incentives contingent upon finishing the questionnaire, such as prize draws for money (Busby and Yoshida 2013; Doerflinger et al. 2010; Dykema et al. 2011; Gajic et al. 2012; Galesic 2006; Göritz 2004b, 2006b, 2008, 2014; Göritz and Luthe 2013a, b, c; O'Neil and Penrod 2001; O'Neil et al. 2003; Porter and Whitcomb 2003a; Sánchez-Fernández et al. 2012; Sauermann and Roach 2013; Wiley et al. 2009), gift cards (Göritz and Luthe 2013b; Porter and Whitcomb 2004; Preece et al. 2010; Wilson et al. 2010; Ziegenfuss et al. 2013), or non-monetary prizes (Cobanoglu and Cobanoglu 2003; Göritz 2004b, 2006a; Göritz and Wolff 2007; Heerwegh 2006; Laguilles et al. 2011; Marcus et al. 2007; McCree-Hale et al. 2010; Ziegenfuss et al. 2013) are very popular among survey designers, empirical findings about their influence on participation behavior in Web surveys are rather inconclusive.

Result summary as incentive Batinic and Moser (2005) argue that offering an automated result report to respondents could increase participation in business-to-business Web surveys where respondents or their businesses might benefit from a summary of the survey outcome. However, several studies show that when targeting consumers in Web surveys, offering a summary report has no effect (Galesic 2006; Göritz 2014; Göritz and Luthe 2013c; Marcus et al. 2007; Sax et al. 2003; Tuten et al. 2004) or even a discouraging effect on sample members (Batinic and Moser 2005; Göritz and Luthe 2013a). Only Edwards et al. (2009) find in their meta-analysis on medical Web surveys that offering survey results increases the odds of participation.

2.3.3 Questionnaire design

The Web questionnaire and all of its characteristics, such as layout, length, and content, are not visible to sample members at the point of the initial survey participation request. Only after clicking on the link in the e-mail invitation or typing the URL into the Web browser, a potential respondent can evaluate these factors and take them into consideration in the participation decision. Therefore page and question specific characteristics, technical issues, and problems with questionnaire design or question wording affect the probability of breakoff during the survey (Peytchev 2009).

Paging vs. scrolling Web questionnaires can be designed in one of two formats. The paging format presents individual or small groups of questions on subsequent screens and respondents submit their answers by clicking the "Next" button. The scrolling format presents the entire Web questionnaire on a single screen and the respondent submits all answers at once at the end of the questionnaire. The paging format allows the questionnaire designer to use more interactive features, such as automated filters and skips based on previous answers, randomization of questions and items, and response format validation (Lozar Manfreda et al. 2002; Peytchev et al. 2006), and to track respondents' behavior on an individual page level (Bosnjak and Tuten 2001; Couper et al. 2001). On the other hand, the scrolling format gives the respondent the opportunity to view the entire Web questionnaire at once and therefore consider length and question content in the participation decision (Lozar Manfreda et al. 2002). Both Lozar Manfreda et al. (2002) and Peytchev et al. (2006) find no influence of the format on break-offs in Web surveys.

Progress indicator Progress indicators provide feedback to respondents about the remaining task when filling out a Web questionnaire in the paging format. Experimental studies find no significant influence of presenting a response indicator on break-offs, for constant progress indicators that show the progress as the number of pages or questions presented to the point divided by the number of total pages or questions in the questionnaire (Healy et al. 2005; Heerwegh and Loosveldt 2006; Yan et al. 2011). For longer Web questionnaires, the use of a progress indicator even shows a negative effect on survey completion (Crawford et al. 2001; Yan et al. 2011). Variation of the speed of the progress indicator results in a decrease of break-offs for fast-to-slow progress indicators, where the progression decelerates across the questionnaire, and an increase for slow-to-fast progress indicators, where the progression speeds up toward the end of the questionnaire, over no indicator (Conrad et al. 2010; Matzat et al. 2009). Villar et al. (2013) confirm these findings in a recent meta-analysis.

Questionnaire layout Waltson et al. (2006) report that the respondents to a pop-up Web survey rate the survey better in terms of attractiveness and stimulation if they receive a "graphic" Web questionnaire that is professionally designed, incorporating various colors, font types, font sizes, and pictures. However, layout features of the Web questionnaire seem to have only very small and mostly non-significant influence on survey completion (Mahon-Haft and Dillman 2010; Wiley et al. 2009; exception: Keusch 2012). Stieger et al. (2011) show that the use of questions that require Java applets in Web surveys increases break-offs. Göritz and Stieger (2008) find that longer loading times for the first page of the Web questionnaire lower the likelihood of finishing the survey.

Questionnaire length Several studies find that the length of a Web questionnaire is positively correlated with break-offs (Göritz 2014; Yan et al. 2011) and negatively correlated with the response rate (Deutskens et al. 2004; Edwards et al. 2009; Galesic and Bosnjak 2009; Ganassali 2008; Göritz 2014; Marcus et al. 2007; exception: Wiley et al. 2009). McCambridge et al. (2011) find no significant effect of the length of a baseline Web questionnaire on response probability to future waves. Several studies also show a strong direct relationship between the announced length of the questionnaire and the probability of starting a Web survey (Crawford et al. 2001; Kaplowitz et al. 2012; Trouteaud 2004; Waltson et al. 2006; Yan et al. 2011; exception: Mavletova et al. 2014). Significantly more respondents break off from Web surveys if the time announced in the invitation misleadingly underestimated the true questionnaire completion time (Crawford et al. 2001; Galesic 2006; Waltson et al. 2006; Yan et al. 2011).

3 Theoretical frameworks explaining survey participation behavior

After reviewing how different factors influence participation in Web surveys, I will next discuss seven theoretical frameworks that have been used to explaining participation behavior in surveys in general and Web surveys in particular. According to Albaum and Smith (2012) the four most cited theories for explaining survey participation behavior are self-perception, cognitive dissonance, commitment/involvement,[1] and social exchange. In addition, several reviews use compliance heuristics, leverage-salience, and planned behavior as theoretical frameworks to explain the survey participation decision (Albaum et al. 1998; Albaum and Smith 2012; Fan and Yan 2010; Han et al. 2009).

3.1 Self-perception theory

Bem's (1972) Self-perception theory (SPT) assumes that people "come to "know" their own attitudes, emotions, and other internal states partially by inferring them from observations of their own overt behavior and/or the circumstances in which this behavior occurs" (p. 5). SPT argues that internal motives instead of external influences drive behavior leading to positive attitudes and self-perception toward their own behavior that influences subsequent behavior. Helgeson et al. (2002) infer that "individuals are more likely to engage in behavior when that behavior is consistent with their views of themselves" (p. 306). Tybout and Yalch (1980) show that individuals who are labeled based on their behavior behave and perceive themselves in a manner consistent with these labels.

A number of authors apply SPT to the survey participation decision process, especially in the context of foot-in-the-door techniques, which involve the use of a small initial request (e.g., answering a few questions via telephone), to increase compliance with a substantial subsequent request (e.g., filling out and sending back a long paper questionnaire) (Allen et al. 1980; Hansen and Robinson 1980; Reingen and Kernan 1977; Tybout and Yalch 1980). However, foot-in-the-door techniques require relatively high administrative effort and costs, counteracting two of the main economic reasons for using Web surveys in the first place, namely low costs and fast data collection. Although Tuten (1997) describes the features of e-mail invitations for Web

[1] Strictly speaking, commitment and involvement are individual determinants of (non)participation in a survey and not a theory per se. However, for the purpose of this paper I will use Albaum and Smith (2012)'s terminology and treat 'commitment/involvement' as equivalent to other, more established theoretical frameworks of survey participation.

surveys, such as the sender and the content of the subject line, as means of "getting a foot in the electronic door" (p. 1), these features do not represent true foot-in-the-door techniques in the sense of SPT, because the sample person has not actively respondent to any request at that point. In line with the findings by Tybout and Yalch (1980), labeling individuals as respondents who regularly participate in Web surveys (e.g., by making a statement about past participation behavior in the invitation to a Web survey in an online panel) could increase the likelihood of participation.

3.2 Cognitive dissonance theory

Cognitive dissonance theory (CDT) developed by Festinger (1957) proposes a model of tension reduction where dissonance is perceived as an unpleasant state that motivates behavior. If two or more cognitive elements (e.g., views, thoughts, opinions, desires, or intentions) are perceived to be dissonant, pressure exists to reduce or eliminate this dissonance. Strategies to reduce dissonance involve changing behavioral or environmental cognitive elements that are dissonant and adding cognitive elements that are consonant.

In the context of surveys, Hackler and Bourgette (1973) suggest to create dissonance in the feelings of potential respondents, "a dissonance that could be resolved by returning the questionnaire to the researcher" (p. 277). Both incentive-based and non-incentive-based techniques can create dissonance when not responding to a survey (Cavusgil and Elvey-Kirk 1998). Furse and Stewart (1984) see CDT as the ideal explanation for survey participation because it is able to integrate much of the empirical literature on inducement techniques within a single model and propose that survey participation is not based on a single decision but rather follows from a set of sequential reactions to features of the survey design. The decision sequence is illustrated in a decision tree for mail surveys (Furse and Stewart 1984) and other survey modes (Albaum and Smith 2012). The idiosyncrasies of online communication call for an adaptation of the model when survey participation requests are sent by e-mail (see Fig. 1).

Web survey sample members who received a new e-mail to their inbox have to decide whether to delete the e-mail without reading or to open it (D_1). Survey design factors that could play a role at this stage of the participation decision process include prenotification messages, the sender of the invitation, and the text of the subject line. Next, a sample member can decide to delete the message without reading or to evaluate the request after opening the e-mail (D_2). Unconditional cash incentives, the most important feature creating cognitive dissonance at this stage in mail surveys (Furse and Stewart 1984), can only be used in Web surveys if sample members are contacted via mail or in person. After evaluating the request to participate in the Web survey, the sample member has three alternative options for how to proceed (D_3): (1) not respond and delete the e-mail, (2) put off the decision, or (3) click on the URL and start the survey. Other than in mail surveys, where the participation request includes a printed questionnaire, the questionnaire itself is of no use in estimating the effort and time needed to participate in the Web survey at this stage because the questionnaire is not visible before the respondent clicks on the link. Hence, for Web surveys the

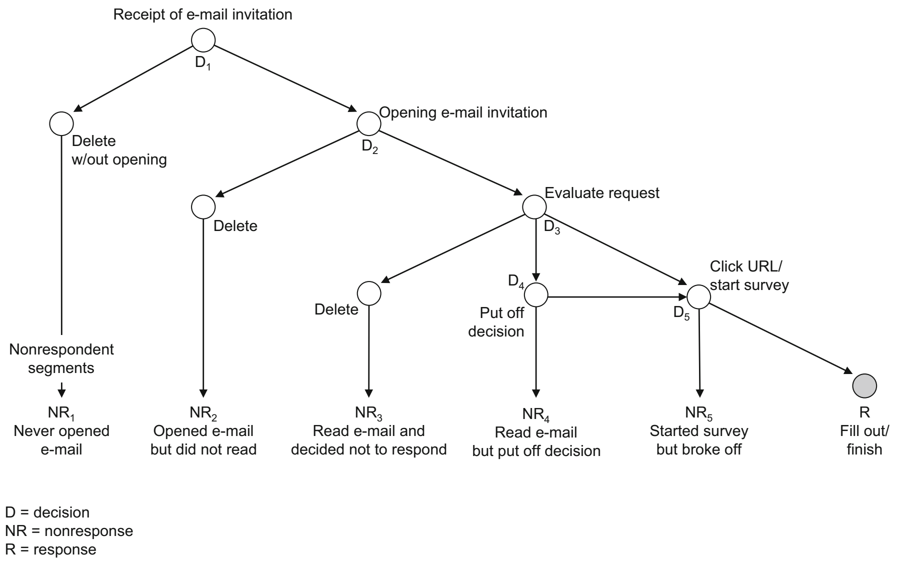

Fig. 1 Response decision process in Web surveys using e-mail invitations [adapted from Furse and Stewart (1984) and Albaum and Smith (2012)]

invitation e-mail plays an even bigger role in the survey participation decision (Greif and Batinic 2007). As with mail surveys, sample members who do put off the decision to participate in the Web survey might be motivated to fill out the questionnaire by means of a reminder message (D_4). The original CDT-based response decision process model for mail surveys proposed by Furse and Stewart (1984) ends at this point. The stage of actually filling out the questionnaire and the associated decision to do so until the last question or to break off is not considered. In Web surveys with a multipage design, researchers have the opportunity to exactly track the behavior of respondents at an individual page level while filling out the questionnaire online. Respondents can further be categorized according to their participation behavior during questionnaire fill out (Bosnjak and Tuten 2001). The Web survey response decision process model proposes that after clicking on the link in the e-mail invitation and starting to fill out the Web questionnaire, respondents can decide to either break off before reaching the final page or to fill out the entire questionnaire (D_5). Peytchev (2009) finds that page and question characteristics (e.g., question content, question type, number of questions, real-time validation) have a stronger influence on Web survey break-off than respondent characteristics.

3.3 Commitment and involvement

The concept of commitment explains why people engage in consistent lines of activity. “Once we make a choice or take a stand, we will encounter personal and interpersonal pressure to behave consistently with that commitment” (Cialdini 2009, p. 5). According to Becker (1960) consistent behavior is characterized as (1) being persistent over some period of time, (2) serving in pursuit of the same goal, and (3) leading to rejection of feasible alternatives. The concept of commitment closely corresponds with what is called enduring involvement or ego involvement in consumer research (Albaum et al. 1998; Muncy and Hunt 1984). Rothshild (1984) defines involvement as "a state of motivation, arousal or interest [...] driven by current external variables (the situation; the product; the communication) and past internal variables (enduring; ego; central values)" (p. 217). Enduring involvement reflects an elevated state of care for an issue, product, or activity that is persistent over a long period of time (e.g., hobbies, personal interests, attitudes) (Havitz and Howard 1995). High involvement is related to strong cognitive and emotional activation leading to intense examination of an issue (Kroeber-Riel and Esch 2004).

Brennan and Hoek (1992) show that enduring involvement with surveys exists and "people tend to respond to mail survey requests in a consistent manner and [...] refusers differ from other nonrespondents with their predisposition toward survey participation" (p. 534). As reported earlier, the same has been demonstrated for Web surveys.

Cavusgil and Elvey-Kirk (1998) also see prenotification and follow-up contacts and the notification of a deadline as potential sources for creating commitment with the survey task by increasing the perceived importance of participating. The influence of prenotification and follow-up messages and the affinity with the sender of the e-mail invitation and the sponsor of the Web survey is well documented.

3.4 Social exchange and benefit-cost theories

Social exchange theory (SET) traces back to the work of Thibaut and Kelly (1959), among others, who describe the consequences of every social interaction as costs and rewards. Rewards are defined as pleasures, satisfaction, and gratification an individual enjoys whereas costs are factors that inhibit or deter the performance of behavior. The higher the burden of an action (e.g., high physical or mental effort necessary, fear or danger of social exposure, conflicting interests), the higher the costs will be.

In the context of surveys, Esser (1986) stresses that the rewards of participating are especially high in situations where participation can be linked to the personal interests of sample members. As for burdens, everyday commodities, such as shortage of time and attractiveness of alternate activities (opportunity costs), and uncertainty about unfamiliar situations, comprehension problems, and fear of loss of anonymity (transaction costs) are factors that drive the costs of survey participation. Dillman (1978) uses SET in his 'Total Design Method', later 'Tailored Design Method' (Dillman 2000; Dillman et al. 2009), to establish the ideal survey research process for different methods of data collection. He extends Thibaut and Kelly's (1959) concept by adding the construct of trust that the promised rewards will in fact be provided.

Singer (2011) proposes a Benefit-cost Theory of survey participation that is closely linked to SET but places a stronger focus on the "inevitable costs that survey participation entails for the respondents" (p. 388). From several studies analyzing open-ended questions about reasons for and against survey participation and using vignettes describing hypothetical surveys, she concludes that concerns about data privacy and confidentiality may especially deter survey participation. Additionally, Helgeson et al. (2002) describe the participation decision process for mail surveys as a hierarchical sequence of individual decisions where sample members ideally move through four stages: (1) attention, (2) intention, (3) completion, and (4) return. While Helgeson et al. (2002) describe the process as being "not a model of the social-exchange process per se", the model still "fits into, and is informed by, social exchange theory" (p. 306). According to the model, an individual decides to participate in a mail survey depending on a cost/benefit evaluation for both survey design factors and respondent factors in subsequent steps.

Researchers use SET in particular to explain the positive influence of small pre-paid incentives on response rates in mail surveys (Groves et al. 1992), which make invited sample members recognize that the request is part of a larger social exchange with the survey organization (Tourangeau 2004). This is in contrast to usually much higher conditional incentives where the request is presented as an economic exchange, in the sense of "If you complete this questionnaire I will pay you for it" (Dillman 2000, p. 168). The lack of an adequate equivalent to unconditional monetary incentives in the online realm was problematized earlier. Directly relating monetary incentives to questionnaire length (e.g., 1 Euro for 10 min), a common practice in online panels, provokes the perception of an economic exchange (Dillman et al. 2009).

Other elements of Web surveys that relate to social exchange suggested by Dillman (2000) and Dillman et al. (2009) that have been proven influential in increasing survey participation are prenotification and follow up messages, i.e., more contacts with the sample persons. Concerning a respondent friendly Web questionnaire, it again has to be considered that the instrument is seen by the respondent only after starting the survey. Therefore page and question specific characteristics, technical problems and problems with questionnaire design or question wording affect the probability of break-off during the survey (Peytchev 2009). Also, not surprisingly, several studies find that the length of a Web questionnaire is negatively correlated with the response rate. Finally, the return process in Web surveys differs from the return process in mail surveys and there is no real equivalent to the prepaid and preaddressed envelop as suggested by Dillman (2000). Nevertheless, as reported earlier, fast and easy access to the Web questionnaire without having to type in an additional password increases participation.

3.5 Compliance heuristics

Social and cognitive psychological research on decision making has contrasted two types of processing (Eagly and Chaiken 1984; Petty and Cacioppo 1986). The first type of processing takes a central route characterized by deep, thorough consideration of relevant cost and benefit related arguments for and against the options. However, decision making about a requested activity based on pertinent arguments, such as the attractiveness of the activity itself, needs time and cognitive effort. In many instances people either lack motivation to take the central route of deep processing or do not have enough detailed information about the task required and will use shallower, quicker decision making that relies on heuristic aspects of options instead (i.e., taking the peripheral route). Based on numerous social psychological experiments and the practical experience of sales and fund-raising organizations and advertising and public-relations agencies, Cialdini (2009) summarizes six fundamental heuristic principles for compliance within society: (1) reciprocity, (2) consistency, (3) social proof, (4) liking, (5) authority, and (6) scarcity. The principle of consistency was already discussed earlier in the context of commitment and involvement. Groves et al. (1992) argue that the compliance heuristics are applicable to the survey participation decision process because potential respondents typically do not have a large interest in survey participation and will not devote large amounts of time or cognitive effort to the participation decision.

Reciprocity is a universal social norm that demands that "(1) people should help those who have helped them, and (2) people should not injure those who have helped them" (Gouldner 1960, p. 171). Individuals, who receive some kind of positive behavior, feel obliged to answer with positive behavior in return trying to repay what another person has provided. The principle of reciprocity acts as an explanation for the numerous studies consistently finding positive effects of unconditional incentives. Cavusgil and Elvey-Kirk (1998) suggest that the obligation to reciprocate and return the questionnaire originates on the receipt of the questionnaire itself because the recipient acknowledges that time, effort, and resources were invested. However, studies on enhancing the Web questionnaire layout to produce a professional image of the survey instrument show inconclusive results.

The principle of social proof builds on the fact that if "objective, non-social means are not available, people evaluate their opinions and abilities by comparison respectively with the opinions and abilities of others" (Festinger 1954, p. 118). An action is deemed appropriate when others are doing it as well, and the more people engage in a behavior, the higher the understanding that this behavior is correct (Cialdini 2009). The concept of social proof is sometimes used by interviewers in the participation request when stressing high response rates in the past. Some unrestricted self-selected Web surveys using nonprobability sampling techniques (Couper 2000), such as the WWW User Surveys (Kehoe and Pitkow 1996; Pitkow and Recker 1994) and the W3B surveys (Fittkau and Maaß 2013), advertise their extremely high number of more than 100,000 participants who took part in previous waves. However, this use of social proof is not unproblematic. First, there is a frequent misconceptualization that larger numbers of participants make nonprobability based samples better (Bethlehem and Stoop 2007). Second, social proof might be an argument for individual sample members not to participate as already many others have done so (Groves et al. 1992).

The principal of liking suggests that people are more likely to be influenced by others they like. Likability is influenced by a number of factors, most dominantly by physical attractiveness and similarity (Cialdini 2009). This heuristic is closely related to situational involvement, defined as a temporarily heightened concern initiated by a specific situation (e.g., a media stimulus, buying process) (Havitz and Howard 1995). Situational involvement can play a decisive role in the survey participation decision because people have different feelings of commitment to the individual aspects of the survey, such as the sponsor, the researcher, the topic, and the research process. Albaum et al. (1998) argue, that the less favorable the attitude toward an aspect of the survey, (1) the lower will be the involvement with/commitment to anything related to the survey, (2) the more destructive survey behavior (e.g., nonresponse, deliberately false answers) will be shown, and (3) the less favorable are the attitudes toward the survey itself. Among situational involvement factors, survey topic seems to play a particularly vital role in the participation decision process in Web surveys (Couper 2005), especially in online panels with declining response rates (Bethlehem and Stoop 2007).

Authority creates strong social pressure to comply with a request. Among others, Milgram's (1974) experiments provided striking evidence for a fundamental sense of duty toward authority, even if the request causes emotional and physical discomfort. Authority-obedience works as a decision-making shortcut when the authority is associated with high levels of knowledge, wisdom, and power (Cialdini 2009, p. 180). The authority-obedience heuristic suggests that survey participation requests coming from individuals and organizations associated with higher authority to collect personal data, such as governmental organizations or universities, are more successful than similar requests that come from people and organizations with lower authority, such as commercial survey organizations (Groves et al. 1992). As previously reported, research on the influence of the authority of the sender of an e-mail invitation confirm the authority heuristic for Web surveys.

The last compliance heuristic deals with the relationship between the availability or scarcity of an item and its perceived value. Objects and opportunities that are scarce or becoming rare are more valuable than items that seem plentiful, even if they are imperfect (Cialdini 2009). Survey researchers can utilize the scarcity heuristic by stressing the scarceness of a participation opportunity (Groves et al. 1992). Survey participation requests that emphasize the chance for respondents to make their voice heard or have their opinion count should have more success when coupled with information that such an opportunity is rare. The limited empirical support for a positive effect of including scarcity statements and deadlines in the Web survey participation request may be explained by the proliferation of surveys in general and Web surveys in particular in Western society in recent years. In general, "people may no longer consider the chance to have their opinions counted as an especially rare, and therefore valuable, event" (Groves et al. 1992, p. 483).

3.6 Leverage-salience theory

Groves et al. (2000) propose the leverage-salience theory (LST) for interviewer administered surveys. LST considers that the importance of different survey attributes can vary among persons. An attribute that is very important for one sample member, acting either as a motivator to participate or a deterrent leading to non-participation, might be absolutely irrelevant for another sample member. "The achieved influence of a particular feature is a function of how important it is to the potential respondent, whether its influence is positive or negative, and how salient it becomes to the sample person during the presentation of the survey request" (p. 301). LST has the following implications (Groves et al. 2009): (1) Sample members use different reasons for accepting or declining a survey request and these reasons are not known to the requestor in advance. (2) No one survey request can address the individual concerns of all sample members. (3) When contacting sample members, interviewers need to identify attributes that are important in the participation decision and make attributes with a positive leverage salient in the survey request. LST helps explaining the roles of incentives, survey topic, and sponsor affinity during the survey request process (Groves et al. 2000, 2004) and is related to cognitive processing and the compliance heuristics. If the topic of a survey is of high personal relevance for a sample member, the central path of extensive cognitive effort is used. On the contrary, sample members seem to process the participation decision on the peripheral path with low cognitive effort if the topic is personally less relevant and therefore use other factors (e.g., incentives, questionnaire length) in the decision process. Because LST was originally proposed to explain participation behavior in interviewer administered survey modes, research on LST in the online realm is very limited (Keusch 2013).

3.7 Theory of planned behavior

Ajzen's (1987,1991) theory of planned behavior (TPB) is based on a model that predicts behavior directly by behavioral intention. According to TPB, intentions capture the motivational factors that have an impact on behavior. Intentions are indicators of how much effort people are willing to put into performing a behavior. Intention to perform a certain behavior is influenced by attitudes toward that behavior, subjective norms, and perceived behavioral control. Attitude is "the amount of affect for or against some object" (Fishbein and Ajzen 1975, p. 11). Subjective norms describe the perceived social pressure to perform or not to perform the behavior. Perceived behavioral control represents the perceived ease or difficulty of a behavior reflected by past experience and anticipated impediments and obstacles. Attitudes toward the behavior and subjective norms depend in turn on behavioral beliefs and normative beliefs, respectively. Belief represents the information a person has about the object.

For Web surveys, Bosnjak et al. (2005) extend the original TPB model by adding the construct of 'moral obligation', defined as "internalized moral rules, not perceptions of others' ideas about what one should do" (p. 495), as a fourth predictor for behavioral intention. In a two-stage study, they show that the four predictors correlate positively with behavioral intention and that the model significantly predicts Web survey participation behavior in a short-term student online panel. In spite of the relatively low predictive power of the model, Bosnjak et al. (2005) see the extended TPB approach as an adequate framework that can explain participation in online panel surveys. Two more recent studies confirm that TPB can satisfactorily explain Web survey participation intention if additional constructs, such as trust, innovativeness (Fang et al. 2009), confidence in decision making, and past behavioral frequency (Haunberger 2011), are added to the model.

4 Expert survey

To assess and quantify the relationship between factors influencing Web survey participation (as described in Sect. 3 above) and theoretical survey participation frameworks (Sect. 4), I conducted an expert survey among psychological researchers. I collected contact information (i.e., names and email addresses) from 477 psychologists with a background in general, experimental, cognitive, consumer, or social psychology from their public profiles available on the websites of psychology departments at universities in Austria, Germany, and Switzerland. All researchers had at least a doctoral degree and no students were contacted for the expert survey. I invited the researchers with a personalized e-mail including a link to a Web questionnaire (see Web Appendix A). Researchers who had not started the expert survey within one week after the initial invitation received a reminder message via e-mail. The Web questionnaire was programmed and hosted in Qualtrics Research Suite and consisted of 11 questions presented on individual screens, each including a short description of one survey participation framework. After each description, respondents had to rate how suited the concept is to explain the influence of 10 factors on Web survey participation using a 6-point, end-labeled rating scale (0 = 'not suited at all' to 5 = 'very well suited'). To shorten the task and to reduce response burden, the questionnaire only included factors that were shown to have a consistent effect on Web survey participation in at least three papers published in peer-reviewed journals. Respondents could choose to fill out the Web questionnaire in German or English (see Web Appendix B). After two weeks of field work, 133 psychological researchers had started the expert survey and 37 had finished the entire questionnaire for a response rate of 7.8 % (AAPOR RR1) and a completion rate of 27.8 % with almost all break-offs occurring on the first two pages of the Web questionnaire. 62 % of the responding researchers are male. The median age of the respondents is 38 years (range 27–64). 84 % work at a university in Germany. Most researchers reported to have their personal research interest in general (46 %), cognitive (41 %), social (38 %), and experimental psychology (35 %).

Table 2 presents the mean ratings for the suitability of the 11 theoretical frameworks to explain the influence of 10 factors on Web survey participation. Bold values over 2.5 indicate that, according to the expert survey, a theoretical framework is above

Table 2 Mean ratings for suitability of theoretical frameworks to explain influence of factors on Web survey participation

	SPT	CDT	C/I	SET	REC	SOP	AUT	LIK	SCA	LST	TPB
Gender	1.4	0.9	1.2	1.2	1.3	1.7	1.1	1.7	0.6	1.8	1.1
Personality	**3.1**	1.8	**2.9**	2.1	2.2	2.1	2.2	2.0	1.0	**3.0**	2.4
Personal topic interest	**3.2**	2.4	**3.5**	**2.9**	1.3	1.4	0.9	2.2	1.5	**3.5**	**3.6**
Attitudes toward survey research	**3.5**	**2.7**	**3.5**	**2.7**	1.7	2.0	1.4	2.1	1.2	**3.0**	**3.8**
Previous participation behavior	**3.7**	**3.0**	**3.8**	2.0	1.8	2.3	1.4	1.5	1.0	2.2	**3.0**
Prenotification	1.3	1.6	2.0	1.9	1.7	1.8	2.3	2.0	1.4	1.8	2.1
Sender/sponsor	1.5	1.9	2.3	**2.6**	2.2	2.2	**4.4**	**3.5**	1.5	2.1	2.3
Reminder	1.6	2.2	**2.5**	2.0	1.8	1.7	**2.7**	1.8	1.3	1.5	2.2
Unconditional incentives	1.8	2.4	2.2	**4.0**	**3.4**	1.4	1.4	1.9	1.9	1.8	2.1
Questionnaire length	1.3	1.5	1.3	**3.1**	1.4	1.0	1.3	0.9	1.2	1.7	2.0

$n = 37$. Scale: 0 = 'not at all suited' to 5 = 'very well suited'. Values over 2.5 printed in bold
SPT self-perception theory, *CDT* cognitive dissonance theory, *C/I* commitment/involvement, *SET* social exchange theory, *REC* reciprocity, *SOP* social proof, *AUT* authority-obedience, *LIK* liking, *SCA* scarcity, *LST* leverage-salience theory, *TPB* theory of planned behavior

average suited to explain the influence of a factor. None of the theoretical frameworks seems particularly suited to explain why women are more likely to participate in Web surveys than men and why prenotification messages sent via offline media (e.g., letter, postcard) increase Web survey participation. In the view of the expert survey participants, the influence of a sample member's personality is best explained by SPT (3.1), and LST (3.0). A number of frameworks, especially TPB (3.6), commitment/involvement (3.5), LST (3.5), and SPT (3.2), seem well suited for explaining the influence of personal topic interest on Web survey participation. TPB (3.8), SPT (3.5), commitment/involvement (3.5), and LST (3.0) are appropriate for explaining why a positive attitude toward surveys in general leads to higher Web survey participation. Commitment/involvement (3.8), SPT (3.6), CDT (3.0), and TPB (3.0) explain why individuals behave consistent in their survey (non-)participation. The positive effect of more familiarity with, trust in, and authority of the sender and sponsor on Web survey participation can best be explained by the compliance heuristics of authority-obedience (4.4) and liking (3.5). According to the ratings from the expert survey, the only two theoretical frameworks that are above average suited to explain why reminders help increase Web survey participation are authority obedience (2.7) and commitment/involvement (2.5). The positive influence of unconditional incentives on Web survey participation is best explained by SET (4.0) and the compliance heuristic of reciprocity (3.4). SET is also best suited to explain why shorter Web questionnaires lead to higher participation (3.1).

To visualize the relationship among the individual theoretical frameworks explaining Web survey participation and between the frameworks and the factors influencing Web survey participation, I conducted a principal component analysis (PCA) on the data in Table 1 using the `prcomp` and `biplot` functions in `R` (R Core Team 2014). According to the expert survey, none of the theoretical frameworks is particularly useful in explaining the influence of sample members' gender and prenotification on Web survey participation nor can the compliance heuristics of social proof and scarcity explain the influence of any of the factors. Hence, these two theoretical frameworks and two factors are not included in the PCA. The first three principal components extracted in the PCA, all with an eigenvalue greater than 1, explain 85.8 % of the total variation in the data. Figure 2 shows the results of the PCA as two biplots with scaled principal components 1 and 2 in the left panel and scaled principal components 1 and 3 in the right panel. A look at the x-axis of the left panel reveals that sample person characteristics, such as attitudes toward survey research, personal topic interest, previous Web survey participation behavior, and sample members personality, load highly negative on principal component 1. On the other hand, survey design characteristics, such as sender and sponsor of the Web survey and the use of a reminder message, have high positive loadings on principal component 1. SPT, commitment/involvement, TPB, and LST can be found on the left side of the biplot, whereas the compliance heuristics of authority-obedience, liking, and reciprocity are all located on the right side of the x-axis. Principal component 2, represented by the y-axis in the left panel of Fig. 2, is defined by high negative loadings of extrinsic motivators, such as unconditional incentives and questionnaire length, on the one side and positive loadings for intrinsic motivators, such as sample member personality, survey sender and sponsor, reminders, and previous participation behavior, on the other side. SET and reciprocity are located

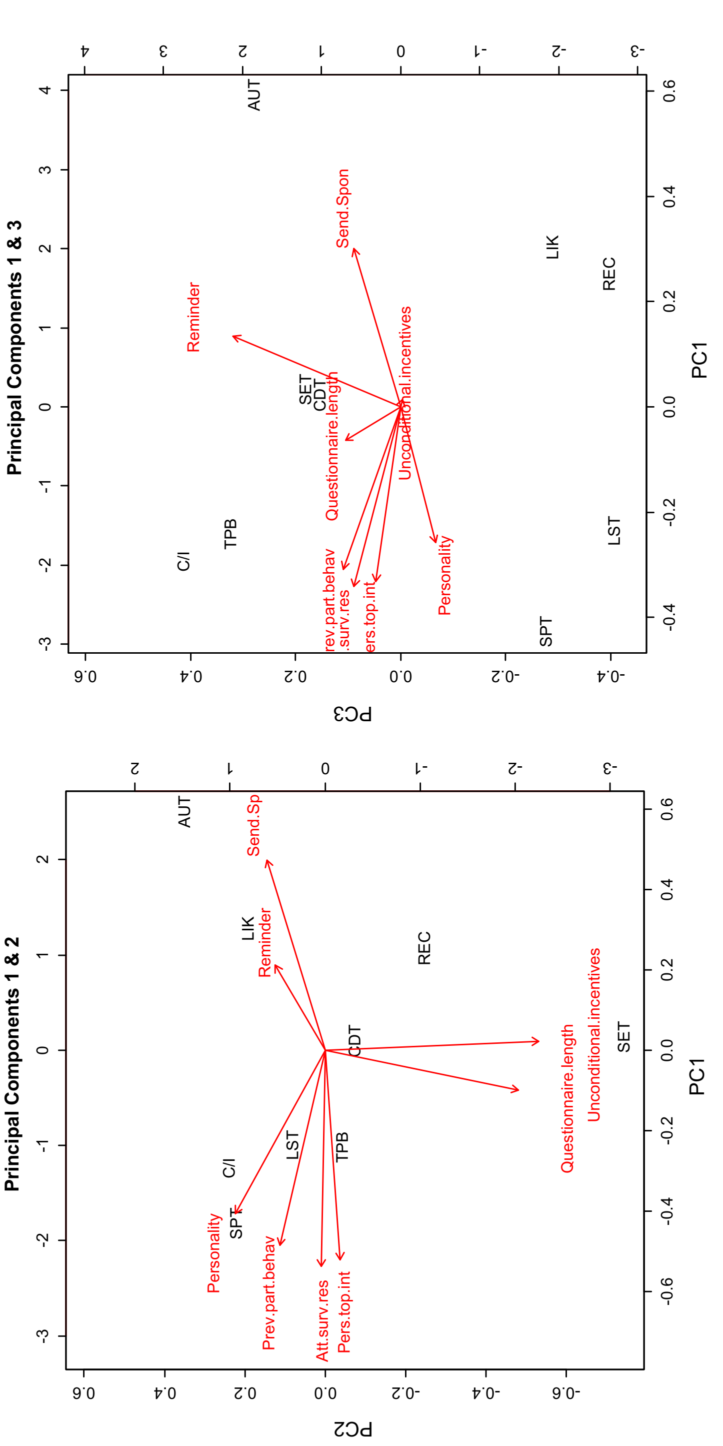

Fig. 2 Biplot of scaled principal components. *SPT* self-perception theory, *CDT* cognitive dissonance theory, *C/I* commitment/involvement, *SET* social exchange theory, *REC* reciprocity, *AUT* authority-obedience, *LIK* liking, *LST* leverage-salience theory, *TPB* theory of planned behavior

close to the negative end of principal component 2 while authority-obedience, commitment/involvement, SPT, and liking are located on the positive side of this axis. Principal component 3, represented by the y-axis in the right panel in Fig. 2, is characterized by high positive loadings of reminder messages and questionnaire length on one side and negative loadings for personality on the other side. Commitment/involvement, TPB, and authority-obedience are located on the positive side of principal component 3 while LST, reciprocity, SPT, liking, and LST are located on the negative side.

5 Discussion

The rapid growth in scientific publications on factors influencing participation behavior in Web surveys underlines the increased interest in Internet survey data collection in market and social research, both in the industry and in academia. This paper presents a systematic review of state-of-the-art literature on participation behavior in Web surveys, updating an earlier review by Fan and Yan (2010). Based on Groves et al. (1992)'s classification of factors influencing participation in self-administered surveys, studies focusing on attributes of the Web survey design and characteristics of the sample person seem to be most prevalent in the literature, while studies analyzing the influence of societal-level factors, such as survey fatigue and cultural characteristics of different populations, are still rare. More research on the effect of these societal-level factors is necessary to better understand the decreasing willingness in the general population to participate in surveys in general and Web surveys in particularly.

In spite of the impressive number of publications in this field, only a handful of survey design attributes and sample person characteristics show consistent influence on Web survey participation behavior, supporting similar observations for traditional survey data collection modes (Helgeson et al. 2002). More systematic research is needed that aims at identifying circumstances under which certain factors, such as contact mode, timing of the invitation, appeals used in the subject line and invitation message, personalization of the invitation, and questionnaire layout, influence Web survey participation and when they do not.

By linking the results of empirical research on factors influencing Web survey participation to existing theoretical frameworks on participation behavior, this paper seeks to improve the understanding about the effect of different factors when inviting people to participate in Web surveys. An expert survey among 37 psychological researchers at Austrian, German, and Swiss universities reveals that most of the theoretical frameworks proposed in the literature for the traditional survey data collection modes are also applicable to Web surveys. Self-perception theory, the concepts of commitment and involvement, leverage-salience theory, theory of planned behavior, and, to some extent, cognitive dissonance theory seem suited to explain the influence of sample member characteristics, such as personality, personal topic interest, attitudes toward survey research, and previous Web survey participation behavior, on the decision to participate in a Web survey. On the other hand, respondents in the expert survey deem some of Cialdini's (2009) compliance heuristics appropriate for explaining the effects of Web survey design attributes, such as the concept of reciprocity explaining the positive influence of unconditional incentives and the concepts of authority-obedience and liking explaining the effect of sender and sponsor and reminder messages. The experts rate Social Exchange Theory as useful in explaining both sample person characteristics, such as personal topic interest and attitudes toward survey research, and Web survey design attributes, such as unconditional incentives and questionnaire length. Surprisingly, Cognitive Dissonance Theory, praised by some survey methodologists as being ideally suited to explain survey participation in other modes (Furse and Stewart 1984), shows only a relative low suitability for explaining the participation decision in Web surveys. Two of the compliance heuristics, social proof and scarcity are seen as not suited to explain any of the factors in the expert survey. According to the expert survey, none of the tested theoretical frameworks can explain why females are more likely to participate in Web surveys than males and why prenotification via offline media, such as postcards and letters, increases Web survey participation. Future research needs to further address this gap in the literature.

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